

# Asymmetric weights break real-analytic functional calculus

A counterexample to Theorem 1.5 of arXiv:2507.16516 as printed

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**Status.** Candidate full counterexample, subject to human verification. For every fixed  $1 \leq q < \infty$ , there is an admissible regular-growth weight  $\omega$  on  $\mathbb{R}$ —also a  $q$ -algebra weight when  $q > 1$ —such that  $A_\omega^q(\mathbb{R})$  is not closed under complex conjugation. Thus the real-analytic map  $F(z) = \bar{z}$ ,  $F(0) = 0$ , refutes Theorem 1.5 under the paper’s explicit definition of a weight. Moreover, closure under conjugation is characterized exactly by  $\omega(x) \asymp \omega(-x)$  almost everywhere.

## 1 Source statement and the missing hypothesis

The source is D. G. Bhimani and K. B. Solanki, *On the weighted Wiener–Lévy theorem: analogue on Euclidean space, strong converse on LCA group and applications to modulation spaces* [1], arXiv:2507.16516v2.

For a locally compact abelian group  $G$ , the paper defines a weight to be a measurable map  $\omega : G \rightarrow [1, \infty)$  satisfying

$$\omega(x + y) \leq \omega(x)\omega(y).$$

It calls  $\omega$  admissible when it satisfies the Beurling–Domar condition

$$\sum_{n \geq 1} \frac{\log \omega(nx)}{n^2} < \infty \quad (x \in G).$$

For  $1 < q < \infty$ , with  $q' = q/(q - 1)$ , a  $q$ -algebra weight additionally satisfies

$$\omega^{-q'} * \omega^{-q'} \leq \omega^{-q'}, \quad \int_G \omega(x)^{-q'} dx < \infty. \quad (1)$$

The exact definition is reproduced in Figure 1. It imposes no symmetry or comparison between  $\omega(x)$  and  $\omega(-x)$ .

On  $\mathbb{R}^d$ , the paper calls a weight of *regular growth* when, for  $r \geq 1$ ,

$$\omega(x/r) \leq \omega(x), \quad \omega(rx) \leq Cr^\alpha \omega(x) \quad (2)$$

for some  $C \geq 1$  and  $\alpha > 0$ . Theorem 1.5 then asserts that if  $\omega$  is admissible, is a  $q$ -algebra weight for  $q > 1$ , and has regular growth, every real analytic  $F$  with  $F(0) = 0$  acts on the corresponding weighted Fourier algebra. Figures 2 and 3 show the statement and its proposed extension.

- (i) A **weight** on  $G$  is a measurable map  $\omega : G \rightarrow [1, \infty)$  which satisfies the submultiplicative condition:

$$\omega(x+y) \leq \omega(x)\omega(y) \quad \text{for all } x, y \in G.$$

- (ii) A weight  $\omega$  on  $G$  is **admissible** if it satisfies the Beurling-Domar (BD) condition

$$\sum_{n \in \mathbb{N}} \frac{\log \omega(nx)}{n^2} < \infty \quad \text{for all } x \in G. \quad (1.1)$$

- (iii) For  $1 < q < \infty$ , a weight  $\omega$  on  $G$  is a  **$q$ -algebra weight** if it satisfies the following conditions

(a) subconvolutive:  $\omega^{-q'} * \omega^{-q'} \leq \omega^{-q'}$ , and

(b)  $\int_G \omega(x)^{-q'} dx < \infty$ ,

where  $q'$  is conjugate of  $q$ , that is,  $\frac{1}{q} + \frac{1}{q'} = 1$ , and  $*$  denotes the convolution.

**Lemma 1.1** (examples [23, 32, 46]).

Figure 1: The paper's definition on source PDF page 3. The crop is taken directly from the archived v2 source PDF [1].

identification  $z \in \mathbb{C} \mapsto z1_{\mathcal{X}} \in \mathcal{X}$ , where  $1_{\mathcal{X}}$  is the identity of  $\mathcal{X}$ . So, if  $\mathcal{X}$  is a unital Banach algebra having  $\mathbb{C}^2$  as its subalgebra, both of which have same identity, then we may replace  $A^q(\Gamma, \mathbb{C}^2)$  by  $A^q(\Gamma, \mathcal{X})$  and the more general result follows using the same techniques presented here.

Now we shall turn our attention to the converse of Theorem 1.4, which is weighted version of Theorem 1.2 for the case of  $G = \mathbb{R}^d = \Gamma$ . First, we state a definition which puts further conditions on weight that is required for our result.

**Definition 1.4.** A weight  $\omega$  on  $\mathbb{R}^d$  is of **regular growth** if it satisfies the following for all  $r \geq 1$  and  $x \in \mathbb{R}^d$

(a)  $\omega\left(\frac{x}{r}\right) \leq \omega(x)$ , and

(b)  $\omega(rx) \leq Cr^\alpha \omega(x)$  for some constant  $C \geq 1$  and  $\alpha > 0$ .

Observe that the polynomial weights  $\omega_s$ , in particular the Japanese weight  $\omega(x) = (1 + |x|^2)^{\frac{s}{2}}$  for  $s > 0$ , are weights of regular growth on  $\mathbb{R}^d$ . Note that if  $\omega$  is a weight of regular growth, then condition (b) implies that  $\omega(x) \leq C|x|^\alpha$  for all  $|x| \geq 1$ , where  $C = \sup_{|x|=1} \omega(x)$ . Refer to [51] for more details on weight of regular growth. We are now ready to state our next main theorem.

**Theorem 1.5.** Let  $1 \leq q < \infty$ ,  $n, d \in \mathbb{N}$ ,  $\omega$  be an admissible ( $q$ -algebra if  $q > 1$ ) weight of regular growth on  $\mathbb{R}^d$ , and let  $\mathcal{X}$  be a unital commutative Banach algebra.

- (i) If  $F : I \rightarrow \mathcal{X}$  ( $I \subset \mathbb{C}$ ) is a real analytic function with  $F(0) = 0$ , then  $T_F$  acts on  $A_\omega^q(\mathbb{R}^d, \mathcal{X})$ .

Figure 2: Regular growth and Theorem 1.5(i), source PDF page 6 [1].

- (ii) If  $F : I \rightarrow \mathbb{C}^n$  ( $I \subset \mathbb{C}^n$ ) is a real analytic function with  $F(0) = 0$ , then  $T_F$  acts on  $A_\omega^q(\mathbb{R}^d, \mathbb{C}^n)$ .

A variant of Theorem 1.5 is obtained in [51, Theorem 1.6.17] for a weight of regular growth and  $q = 1$ . Feichtinger, Kobayashi and Sato [27, Theorem 1.5] extended it for  $1 \leq q < \infty$  with Japanese weight  $\omega_s$  (cf. (5.6) below and [26, Theorem 5.1]). It is worth noting the key difference between these results and Theorem 1.5: In [26, 27], the considered functions are in  $f \in A_\omega^1(\mathbb{R}^d)$  whose range is contained in a compact subset of  $\mathbb{R}^d$  with  $\omega$  a weight of regular growth or real valued functions with Japanese weight  $f \in A_{\omega_s}^q(\mathbb{R}^d)$ , and  $F$  being analytic on a neighborhood of  $f(\mathbb{R}^d) \cup \{0\}$ . While Theorem 1.5 encompasses all  $f \in A_\omega^q(\mathbb{R}^d)$  whose range lies in the domain of a real analytic function  $F$  and  $\omega$  is a weight of regular growth.

**Remark 1.4.** *We do not know whether Theorem 1.5 can be extended to arbitrary  $\Gamma$  and not only  $\mathbb{R}^d$  as the techniques presented here used the scaling property in  $\mathbb{R}^d$ . And it is the reason we consider the weights of regular growth which behaves “nicely” with scaling.*

### 1.3. Applications to time-frequency spaces.

Figure 3: Theorem 1.5(ii) and Remark 1.4 asking about arbitrary  $\Gamma$ , source PDF page 7 [1].

For a weight on  $\mathbb{R}^d$ , write

$$L_\omega^q = \left\{ g : \int_{\mathbb{R}^d} |g(x)|^q \omega(x)^q dx < \infty \right\}, \quad A_\omega^q = \mathcal{F}(L_\omega^q).$$

Under the source hypotheses,  $L_\omega^q \subset L^1$ : this is immediate for  $q = 1$  because  $\omega \geq 1$ , and for  $q > 1$  follows from Hölder's inequality and the second condition in (1). Thus the Fourier transform is defined on  $L_\omega^q$  and is injective there.

## 2 Exact characterization of the obstruction

The proof intuition is short. Complex conjugation in physical space reflects and conjugates the inverse Fourier datum:

$$\widehat{\bar{g}} = \widehat{Jg}, \quad (Jg)(x) = \overline{g(-x)}. \quad (3)$$

The weighted norm of  $Jg$  sees  $\omega(-x)$  rather than  $\omega(x)$ . Consequently, conjugation preserves the Fourier algebra precisely when the weight cannot distinguish the two half-spaces by an unbounded factor.

**Theorem 1** (Conjugation characterization). *Let  $1 \leq q < \infty$  and let  $\omega : \mathbb{R}^d \rightarrow [1, \infty)$  be a measurable weight such that  $L_\omega^q \subset L^1$ . The following are equivalent:*

- (i)  $f \in A_\omega^q$  implies  $\bar{f} \in A_\omega^q$ ;
- (ii)  $Jg = \overline{g(-\cdot)}$  maps  $L_\omega^q$  into itself;
- (iii) there is  $M < \infty$  such that

$$M^{-1}\omega(x) \leq \omega(-x) \leq M\omega(x) \quad \text{for almost every } x.$$

When these conditions hold, the operator norm of  $J$  is  $\text{ess sup}_x \omega(-x)/\omega(x)$ .

*Proof.* Identity (3) and Fourier injectivity on  $L^1$  give the equivalence of (i) and (ii). Indeed, if  $\widehat{\bar{g}} = \widehat{h}$  for some  $h \in L_\omega^q$ , then  $\widehat{h} = \widehat{Jg}$  and hence  $h = Jg$ .

If  $J$  maps all of  $L_\omega^q$  into itself, its graph is closed. To see this, suppose  $g_n \rightarrow g$  and  $Jg_n \rightarrow h$  in  $L_\omega^q$ . Since  $\omega \geq 1$ , both convergences also hold in unweighted  $L^q$ ; reflection is an isometry there, so  $Jg_n \rightarrow Jg$  and  $h = Jg$ . The closed graph theorem makes  $J$  bounded.

A change of variables gives the exact identity

$$\|Jg\|_{L_\omega^q}^q = \int_{\mathbb{R}^d} |g(x)|^q \omega(-x)^q dx. \quad (4)$$

Thus  $J$  is bounded if and only if  $\omega(-x)/\omega(x)$  is essentially bounded. The necessity follows, for example, by testing characteristic functions on finite-measure subsets of its level sets; sufficiency follows directly from (4). Replacing  $x$  by  $-x$  supplies the reverse inequality. Formula (4) also yields the stated norm.  $\square$

Since  $F(z) = \bar{z}$  is real analytic in the paper's explicit real-variable sense, Theorem 1 shows that reflection comparability is a *necessary* hypothesis for the universal real-analytic conclusion in Theorem 1.5. We next show that none of the other printed conditions forces it.

### 3 Asymmetric weights satisfying every printed condition

Fix real numbers  $a > b \geq 0$  and define

$$w_0(x) = \begin{cases} (1+x)^a, & x \geq 0, \\ (1+|x|)^b, & x < 0. \end{cases} \quad (5)$$

**Lemma 2.** *The function  $w_0$  is a submultiplicative weight, has regular growth, and satisfies the Beurling–Domar condition. Moreover,  $w_0(x)/w_0(-x) = (1+x)^{a-b}$  for  $x > 0$ , so it is not reflection comparable. These properties, except for the displayed normalization at zero, persist after multiplication by any constant  $L \geq 1$ .*

*Proof.* For two arguments of the same sign, submultiplicativity follows from  $1+s+t \leq (1+s)(1+t)$  and the appropriate exponent. If  $x \geq 0 > y$ , then: when  $x+y \geq 0$ ,

$$w_0(x+y) = (1+x-|y|)^a \leq (1+x)^a \leq w_0(x)w_0(y),$$

while when  $x+y < 0$ ,

$$w_0(x+y) = (1+|y|-x)^b \leq (1+|y|)^b \leq w_0(x)w_0(y).$$

The other mixed order is identical.

Positive dilation does not change the half-line. Monotonicity gives  $w_0(x/r) \leq w_0(x)$ , and, since  $a > b$ ,  $w_0(rx) \leq r^a w_0(x)$  for  $r \geq 1$ . Thus (2) holds. Finally,

$$\log w_0(nx) \leq a \log(1+n|x|),$$

and the series of this expression divided by  $n^2$  converges. Multiplying by  $L \geq 1$  adds  $\log L$  to the Beurling–Domar estimate and gives

$$Lw_0(x+y) \leq Lw_0(x)w_0(y) \leq (Lw_0(x))(Lw_0(y)),$$

while leaving the other arguments unchanged. □

For  $q = 1$ , Lemma 2 already supplies all the required weight hypotheses; one may take  $a = 1$ ,  $b = 0$ , and  $\omega = w_0$ . For  $q > 1$ , the paper also requires the normalized subconvolution inequality.

**Lemma 3** (Exact  $q$ -algebra normalization). *Let  $1 < q < \infty$ , put  $r = q'$ , and choose  $a > b > 1/r$ . Set*

$$\begin{aligned} C_+ &= \frac{2}{br-1} + \frac{2^{ar+1}}{ar-1}, \\ C_- &= \frac{2}{ar-1} + \frac{2^{br+1}}{br-1}, \quad C = \max\{C_+, C_-\}. \end{aligned}$$

*If  $L \geq \max\{1, C^{1/r}\}$  and  $\omega = Lw_0$ , then  $\omega$  is an admissible regular-growth  $q$ -algebra weight in the exact sense of the paper.*

*Proof.* Put  $v_0 = w_0^{-r}$ . Since  $ar > 1$  and  $br > 1$ ,

$$\int_{\mathbb{R}} v_0(x) dx = \frac{1}{ar-1} + \frac{1}{br-1} < \infty.$$

We prove  $v_0 * v_0 \leq Cv_0$ . Let  $x \geq 0$ . On  $y < 0$  we have  $v_0(x-y) \leq v_0(x)$ , while on  $y > x$  we have  $v_0(y) \leq v_0(x)$ . Each tail therefore contributes at most  $v_0(x)/(br-1)$ . On  $0 \leq y \leq x$ , split at  $x/2$ . On

the first half,  $v_0(x - y) \leq 2^{ar} v_0(x)$ ; on the second the analogous estimate holds for  $v_0(y)$ . Integrating the remaining factor over a half-line bounds the middle contribution by

$$\frac{2^{ar+1}}{ar-1} v_0(x).$$

Hence  $(v_0 * v_0)(x) \leq C_+ v_0(x)$ . For  $x < 0$ , the same three-interval argument with  $a$  and  $b$  interchanged gives  $(v_0 * v_0)(x) \leq C_- v_0(x)$ .

Now  $v = \omega^{-r} = L^{-r} v_0$ , so

$$v * v = L^{-2r} (v_0 * v_0) \leq CL^{-2r} v_0 = CL^{-r} v \leq v.$$

Together with integrability, this is exactly (1). Admissibility and regular growth follow from Lemma 2.  $\square$

## 4 Counterexample for every finite $q$

**Theorem 4** (Full counterexample family). *For every fixed  $1 \leq q < \infty$ , there exist a weight  $\omega$  satisfying all the hypotheses of Theorem 1.5 in dimension  $d = 1$ , and an  $f \in A_\omega^q(\mathbb{R})$ , such that*

$$F \circ f = \bar{f} \notin A_\omega^q(\mathbb{R}) \quad \text{for } F(z) = \bar{z}.$$

Consequently both parts of Theorem 1.5 are false as printed, already for the scalar algebra  $\mathcal{X} = \mathbb{C}$  and, in the multivariate part,  $n = 1$ .

*Proof.* For  $q = 1$ , take  $a = 1$ ,  $b = 0$  and  $\omega = w_0$ . For  $q > 1$ , choose any  $a > b > 1/q'$  and take the scaled weight supplied by Lemma 3. In either case put

$$s = \frac{a+b}{2} + \frac{1}{q}, \quad g(x) = \mathbf{1}_{(-\infty, -1)}(x) |x|^{-s}.$$

On the negative half-line,

$$q(s-b) = 1 + \frac{q(a-b)}{2} > 1,$$

and therefore

$$\int_{\mathbb{R}} |g(x)|^q \omega(x)^q dx < \infty.$$

Thus  $g \in L_\omega^q$ . On the other hand,

$$(Jg)(x) = \mathbf{1}_{(1, \infty)}(x) x^{-s}, \quad q(s-a) = 1 - \frac{q(a-b)}{2} < 1,$$

so

$$\int_{\mathbb{R}} |Jg(x)|^q \omega(x)^q dx = \infty.$$

Let  $f = \widehat{g}$ . Then  $f \in A_\omega^q$ , whereas  $\bar{f} = \widehat{Jg}$  cannot belong to  $A_\omega^q$  by Fourier injectivity, exactly as in Theorem 1.

Finally,  $F(z) = \bar{z}$  is a polynomial in the real variables  $z = x + iy$ , namely  $F(x, y) = x - iy$ . It is therefore real analytic under the paper's Definition 1.2 and satisfies  $F(0) = 0$ .  $\square$

## 5 Where the source proof fails

The source proof begins with  $f = f_1 + if_2$  and says that  $f_1$  and  $f_2$  belong to  $A_\omega^q$  because they are the real and imaginary parts of  $f$ ; see Figure 4. But

$$\operatorname{Re} f = \frac{f + \bar{f}}{2}, \quad \operatorname{Im} f = \frac{f - \bar{f}}{2i},$$

so that step requires closure under conjugation. Theorem 1 proves that this is equivalent to the absent hypothesis  $\omega(x) \asymp \omega(-x)$ . The counterexample makes the first of these real/imaginary parts fail to lie in the algebra.

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whenever the series converges. Note that both  $f_1$  and  $f_2$  are in  $A_\omega^q(\mathbb{R}^d)$ , being the real and imaginary part of  $f$ .

Let  $\phi \in A_\omega^q(\mathbb{R}^d)$  be such that  $\phi \equiv 1$  in some neighborhood  $V$  of 0. For  $\lambda > 0$ , define

$$g_\lambda(t) = [f(\lambda t + t_0) - f(t_0)] \phi(t) \quad (t \in \mathbb{R}^d). \quad (5.7)$$

Then  $\widehat{g_\lambda} = \widehat{R_{t_0} f_\lambda} * \widehat{\phi} - (R_{t_0} f)(0) \widehat{\phi}$ , where  $(R_{t_0} f)(t) = f(t + t_0)$  and  $f_\lambda(t) = f(\lambda t)$  for all  $t \in \mathbb{R}^d$  and from Lemma 5.2 it follows that that  $g_\lambda \in A_\omega^q(\mathbb{R}^d)$  and  $\|g_\lambda\|_{A_\omega^q} < \delta$  for some  $\lambda$  near to 0 and we now fix this  $\lambda$ . Now, writing  $g_\lambda = g_{\lambda,1} + ig_{\lambda,2}$ , see that the function

$$h(t) = F(z_0) \phi(t) + \sum_{(m,n) \in \mathbb{N}_0^2 \setminus \{(0,0)\}} c_{mn} g_{\lambda,1}(t)^m g_{\lambda,2}(t)^n \quad (5.8)$$

belongs to  $A_\omega^q(\mathbb{R}^d, \mathcal{X})$  as the series on right side converges absolutely because  $\|g_{\lambda,j}\|_{A_\omega^q} \leq \|g_\lambda\|_{A_\omega^q} < \delta$  for  $j = 1, 2$ . So, if we set  $g_{t_0} = h\left(\frac{t - t_0}{\lambda}\right)$ . Then  $g_{t_0} \in A_\omega^q(\mathbb{R}^d, \mathcal{X})$  and  $g_{t_0}(t) = F(f(t))$  in  $V = \{t : t - t_0 \in V\}$  which is an open neighborhood of  $t_0$ . Thus  $F(f)$

Figure 4: The unsupported real/imaginary-part step at the top of source PDF page 26 [1].

The exact statement correction forced by the advertised universal real-analytic calculus is therefore:

Add at least the hypothesis  $\omega(x) \asymp \omega(-x)$  almost everywhere.

This correction is necessary and sufficient for the conjugation step. We do *not* claim here that it is by itself sufficient to validate every other localization and scaling step in the proof of Theorem 1.5.

Remark 1.4's proposed extension to arbitrary  $\Gamma$  also cannot hold under the same printed hypotheses: the Euclidean base statement already fails. This does not settle the paper's separate question of replacing the Beurling–Domar condition by the Gelfand–Raikov–Shilov condition.

## 6 Verification, interpretation, and bounded novelty

The construction is self-contained. Its only analytic input beyond elementary integration is Fourier uniqueness on  $L^1$  and the closed graph theorem. A companion script checks representative parameters for  $q = 1, 2, 3$ , evaluates the constants in Lemma 3, runs deterministic random submultiplicativity tests, and numerically samples the convolution ratio. No numerical computation is used in the proof.

The terminology caveat is concrete. Some authors define a “weight” in a particular involutive weighted algebra to include symmetry. Here, however, the source gives an explicit three-part definition,

and neither that definition nor the regular-growth definition contains symmetry. If the authors intended a symmetric weight by convention, this packet identifies a precise missing hypothesis and proof dependency rather than refuting the intended symmetric statement.

A bounded search through 9 August 2026 checked the current arXiv v2 record, the exact title and arXiv identifier together with “counterexample”, “correction”, and “erratum”, and searches combining weighted Fourier or Beurling algebras with conjugation, involution, and reflection symmetry. Run-local indexes were also checked. No correction or prior statement of this exact counterexample was found. General reflection-symmetry issues in weighted algebras are familiar, so novelty is claimed only for the explicit application to Theorem 1.5 and the all- $q$  construction; an exhaustive bibliographic review was not performed.

**Human-review recommendation.** Check the intended convention for “weight” with the authors. As a mathematical statement under the definition printed on page 3, Theorem 1.5 is false for every finite  $q$ , and Theorem 1 pinpoints the exact missing condition for the first indispensable real-analytic operation.

## References

- [1] D. G. Bhimani and K. B. Solanki, *On the weighted Wiener–Lévy theorem: analogue on Euclidean space, strong converse on LCA group and applications to modulation spaces*, arXiv:2507.16516v2 (2025).